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EXAMINER	
HUYNH, CHRISTINE NGUYEN	
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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

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Office Action Summary	Application No. 19/364,868	Applicant(s) Fang et al.	
	Examiner Christine N Huynh	Art Unit 3662	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 25 October 2025.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-19 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-19 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 25 October 2025 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Status of Claims

- This action is in reply to the patent application filed on October 21, 2025.
- Claims 1-19 are currently pending and have been examined.
- This action is made Non-FINAL.
- The examiner would like to note that this application is being handled by examiner Christine Huynh.

Claim Rejections - 35 USC § 101

2. 35 U.S.C. 101 reads as follows:

> Whoever invents or discovers any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof, may obtain a patent therefor, subject to the conditions and requirements of this title.

3. Claims 1-19 rejected under 35 U.S.C. 101 because the claimed invention is directed to an abstract idea without significantly more.

101 Analysis – Step 1

Claim 1 is directed towards an apparatus and claim 14 is directed towards a system. Therefore, claims 1 and 14 are within at least one of the four statutory categories.

101 Analysis – Step 2A, Prong I

Regarding Prong I of the Step 2A analysis in the 2019 PEG, the claims are to be analyzed to determine whether they recite subject matter that falls within one of the follow groups of abstract

ideas: a) mathematical concepts, b) certain methods of organizing human activity, and/or c) mental processes.

Independent claim 1 includes limitations that recite an abstract idea (emphasized below). Claim 1 recites:

1. An apparatus, comprising:

- a policy acquisition circuit configured to **interpret a vehicle policy data value including an external event trigger condition**;
- a policy processing circuit configured to **generate parsed policy data including the external event trigger condition**; and
- a policy execution circuit configured to collect vehicle data from one or more vehicle end points in response to the parsed policy data.

Independent claim 14 includes limitations that recite an abstract idea (emphasized below). Claim 14 recites:

14. A cloud system, comprising:

- a request interface configured to **interpret at least one vehicle data collection description and a target vehicle description**;
- a policy creator circuit configured to **determine at least one vehicle policy data value in response to the at least one vehicle data collection description and the target vehicle description**;
- a policy implementation circuit configured to **determine a group of vehicles in response to the target vehicle description,** and to provide at least one of the at least one vehicle policy data values to the group of vehicles; and
- a cloud interface configured to receive at least one of:

- at least a portion of identified vehicle data in response to the at least one vehicle policy data value; or
- an alert response value in response to the at least one vehicle policy data value.

The examiner submits that the foregoing bolded limitation(s) constitute a "mental process" because under its broadest reasonable interpretation, the claim covers performance of the limitation in the human mind. For example, "**interpret a vehicle policy data value including an external event trigger condition**", "**generate parsed policy data including the external event trigger condition", "interpret at least one vehicle data collection description and a target vehicle description**", "**determine at least one vehicle policy data value in response to the at least one vehicle data collection description and the target vehicle description**", and "**determine a group of vehicles in response to the target vehicle description**..." in the context of this claim encompasses a person (user) looking at data collected and forming a simple judgement. For example, when given vehicle policy data value, determining or calculating what the trigger condition would be. A person can also understand, and parse apart vehicle information given and make a judgment based on the vehicle policy data value based on the vehicle information. If given a plurality of vehicles and vehicle information, a person could determine which vehicles of a plurality of vehicles correlates to the information necessary. Accordingly, the claim recites at least one abstract idea.

101 Analysis – Step 2A, Prong II

Regarding Prong II of the Step 2A analysis in the 2019 PEG, the claims are to be analyzed to determine whether the claim, as a whole, integrates the abstract idea into a practical application. As noted in the 2019 PEG, it must be determined whether any additional elements in the claim beyond the abstract idea integrate the exception into a practical application in a manner that imposes a meaningful limit on the judicial exception. The courts have indicated that additional elements merely using a computer to implement an abstract idea, adding insignificant extra solution activity, or generally linking

use of a judicial exception to a particular technological environment or field of use do not integrate a judicial exception into a "practical application."

In the present case, the additional limitations beyond the above-noted abstract idea are as follows (where the underlined portions are the "additional limitations" while the bolded portions continue to represent the "abstract idea"):

1. An apparatus, comprising:

- a policy acquisition circuit configured to **interpret a vehicle policy data value including an external event trigger condition**;
- a policy processing circuit configured to **generate parsed policy data including the external event trigger condition**; and
- a policy execution circuit configured to collect vehicle data from one or more vehicle end points in response to the parsed policy data.

14. A cloud system, comprising:

- a request interface configured to **interpret at least one vehicle data collection description and a target vehicle description**;
- a policy creator circuit configured to **determine at least one vehicle policy data value in response to the at least one vehicle data collection description and the target vehicle description**;
- a policy implementation circuit configured to **determine a group of vehicles in response to the target vehicle description,** and to provide at least one of the at least one vehicle policy data values to the group of vehicles; and
- a cloud interface configured to receive at least one of:
- at least a portion of identified vehicle data in response to the at least one vehicle policy data value; or

- an alert response value in response to the at least one vehicle policy data value.

For the following reason(s), the examiner submits that the above identified additional limitations do not integrate the above-noted abstract idea into a practical application.

Regarding the additional limitations of "collect vehicle data from one or more vehicle end points…", "provide at least one of the at least one vehicle policy data values to the group of vehicles…", …to receive at least one of: at least a portion of identified vehicle data …", and "to receive at least one of: an alert response value…", the examiner submits that these limitations are insignificant extra-solution activities that merely use a computer (circuit) to perform the process. In particular, the collecting vehicle data steps from the sensors and from the external source are recited at a high level of generality (i.e. as a general means of gathering vehicle and road condition data for use in the evaluating step), and amounts to mere data gathering, which is a form of insignificant extra-solution activity. The providing vehicle policy data values step is also recited at a high level of generality (i.e. as a general means of sending the calculated vehicle policy data values from the generating step), and amounts to mere data transmission, which is a form of insignificant extra-solution activity. The receiving a portion of identified vehicle data or an alert response value step are recited at a high level of generality (i.e. as a general means of receiving calculated vehicle data or a calculated value), and amounts to mere data receiving which is a form of insignificant extra-solution activity. Lastly, the "circuits" are recited at a high-level of generality (i.e., as a generic processor performing a generic computer function of ranking information based on a determined amount of use) such that it amounts no more than mere instructions to apply the exception using a generic computer component.

Thus, taken alone, the additional elements do not integrate the abstract idea into a practical application. Further, looking at the additional limitation(s) as an ordered combination or as a whole, the limitation(s) add nothing that is not already present when looking at the elements taken individually. For instance, there is no indication that the additional elements, when considered as a whole, reflect an

improvement in the functioning of a computer or an improvement to another technology or technical field, apply or use the above-noted judicial exception to effect a particular treatment or prophylaxis for a disease or medical condition, implement/use the above-noted judicial exception with a particular machine or manufacture that is integral to the claim, effect a transformation or reduction of a particular article to a different state or thing, or apply or use the judicial exception in some other meaningful way beyond generally linking the use of the judicial exception to a particular technological environment, such that the claim as a whole is not more than a drafting effort designed to monopolize the exception (MPEP § 2106.05). Accordingly, the additional limitation(s) do/does not integrate the abstract idea into a practical application because it does not impose any meaningful limits on practicing the abstract idea.

101 Analysis – Step 2B

Regarding Step 2B of the Revised Guidance, representative independent claim 1 does not include additional elements (considered both individually and as an ordered combination) that are sufficient to amount to significantly more than the judicial exception for the same reasons to those discussed above with respect to determining that the claim does not integrate the abstract idea into a practical application. As discussed above with respect to integration of the abstract idea into a practical application, the additional element of using a vehicle controller to perform the evaluating... amounts to nothing more than mere instructions to apply the exception using a generic computer component. Mere instructions to apply an exception using a generic computer component cannot provide an inventive concept. And as discussed above, the additional limitations of "collect vehicle data from one or more vehicle end points...", "provide at least one of the at least one vehicle policy data values to the group of vehicles...", ...to receive at least one of: at least a portion of identified vehicle data ...", and "to receive at least one of: an alert response value...", the examiner submits that these limitations are insignificant extra-solution activities. Hence, the claim is not patent eligible.

Dependent claim(s) 2-13 and 15-19 do not recite any further limitations that cause the claims to be directed towards statutory subject matter. The claims merely recite: [repeat the judicial exception]. Each of the further limitations expound upon the [repeat judicial exception] and do not recite additional elements integrating the [repeat judicial exception] into a practical application or additional elements that are not well-understood, routine or conventional. Therefore, dependent claims 2-13 and 15-19 are similarly rejected as being directed towards non-statutory subject matter.

Therefore, claim(s) 1-19 is/are ineligible under 35 USC §101.

Claim Rejections - 35 USC § 102

4. In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

5. The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –
>
> (a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale, or otherwise available to the public before the effective filing date of the claimed invention.

6. **Claim(s) 1-7 and 13-17** is/are rejected under 35 U.S.C. 102(a)(1) as being anticipated by Tsuyunashi et al. (US 20200077292 A1).

Regarding claims 1-7 and 13-17:

With respect to claim 1, Tsuyunashi teaches:

- **a policy acquisition circuit configured to interpret a vehicle policy data value including an external event trigger condition**; ("The data user sets a data collection condition (user data collection request) using the user terminal 10 connected to the data collection apparatus 1. The data collection apparatus 1 creates tag data generation data for generating tag data having index-data properties. The tag data is associated with real data to be collected and is used to search for the data and grasp the outline of the data." [0046]), where a vehicle policy can be set, and therefore a data value such as the tag data including an external event trigger condition can be determined.
- **a policy processing circuit configured to generate parsed policy data including the external event trigger condition**; ("The tag data generation data is generated based on an operation performed by the data user using a computer program and generation data stored in the user terminal 10 or the data collection apparatus 1. The data collection condition and the tag data generation data are stored in the data collection apparatus 1. The data collection condition and the tag data generation data are transmitted to data collection object vehicles (vehicle condition of which is specified by the data user) and also stored in the on-vehicle device 50." [0047]), where once a vehicle policy can be set, an external event trigger condition can be determined.
- **a policy execution circuit configured to collect vehicle data from one or more vehicle end points in response to the parsed policy data**; ("Subsequently, the on-vehicle device 50 each monitor output data from the sensors and the camera. If an event that satisfies the stored data collection condition occurs, the on-vehicle device 50 each store real data R in a storage device. The on-vehicle device 50 each generate and store therein tag data T corresponding to the real data R (vehicle state data) based on the real data R and the stored tag data generation data." [0048]), where the vehicle data is collected based on the data collection condition. When the data collection condition occurs, then vehicle data is collected.

With respect to claim 2, Tsuyunashi discloses the limitations of claim 1. Tsuyunashi teaches data collection of claim 1. Tsuyunashi further teaches:

- **wherein the external event trigger condition comprises a traffic condition value**; ("Because the data amount of the tag data T is small, the present embodiment may store all the pieces of tag data T that match the data collection condition in the data collection apparatus 1. If the road sections specified for data collection are significantly different in traffic volume, the present embodiment may generate, transmit, and accumulate the tag data T intermittently for the road section with a larger traffic volume and delete old data when the collection amount exceeds an appropriate collection amount, for example. Furthermore, the present embodiment may ease the data collection condition for the road section with a significantly small traffic volume and complementarily collect similar data, for example." [0089], "If a fire or a traffic accident occurs near the vehicles, for example, the data collection apparatus 1A transmits a transmission request for requesting camera video from the on-vehicle device 50A positioned near the scene. The data collection apparatus 1A provides the camera video to the emergency vehicles, thereby enabling the emergency vehicles to check the state of the scene before arriving at the scene." [0187]), where the trigger condition for collecting data can include traffic conditions including traffic volume and traffic accidents.

With respect to claim 4, Tsuyunashi discloses the limitations of claim 1. Tsuyunashi teaches data collection of claim 1. Tsuyunashi further teaches:

- **wherein the external event trigger condition comprises a road condition value**; "The time when the vehicles travel and their positional information are transmitted to the data collection apparatus 1 as the tag data T, for example. If the developer needs image data on a certain road,

the developer refers to the tag data T in the data collection apparatus 1 to identify the vehicle that has passed on the object road. The developer can acquire the image data from the vehicle via the data collection apparatus 1." [0087], "The data collection apparatus 1D analyses the collected travel data, thereby grasping the behaviors of the vehicles relating to the congestion in detail. The data collection apparatus 1D determines the travel lanes on which the respective vehicles should travel based on the collected travel data (Step S4). The data collection apparatus 1D, for example, can determine the travel lanes on which the respective vehicles should travel such that the congestion is alleviated or prevented from occurring on the travel lanes. The data collection apparatus 1D notifies the vehicles of travel lane information on the travel lanes determined for the respective vehicles (Step S5)." [0417-0418], "The determiner 33D can grasp the behaviors of the respective vehicles based on the travel data and determine the travel lanes on which the respective vehicles should travel such that congestion is alleviated or prevented from occurring on the travel lanes. The determiner 33D, for example, can instruct vehicles traveling on an expressway (in particular, expressway where congestion occurs) to change the travel lanes appropriately for the lanes on which they are traveling." [0455]), where collecting data can be based on the road condition such as information regarding the road segment or the lanes in the road.

With respect to claim 5, Tsuyunashi discloses the limitations of claim 1. Tsuyunashi teaches data collection of claim 1. Tsuyunashi further teaches:

- **wherein the policy execution circuit is further configured to collect vehicle data in response to an authorization value associated with the vehicle policy data value**; ("As illustrated in FIG. 9, a collection condition table 42a is information having "user ID", "request ID", and "collection conditions" in a manner associated with one another. The "user ID" is an identifier for

identifying a user. The "request ID" is an identifier for identifying a collection request. The "collection conditions" are information indicating collection conditions for real data. The "collection conditions" include "object vehicle condition", "recording trigger", and "collection contents", for example." [0128-0129]), where a user ID can be used for identifying authorization regarding collecting vehicle data. Thus, it would have been obvious to a person of ordinary skill in the art where a user ID provides authorization in an attempt to provide an improved system or method, as a person with ordinary skill has good reason to pursue the known options within his or her technical grasp, and user ID indicates an identification system. In turn, because the product as claimed has the properties predicted by the prior art, it would have been obvious to make the system or product where the vehicle data is collected in response to an authorization value, which is a user ID.

With respect to claim 6, Tsuyunashi discloses the limitations of claim 5. Tsuyunashi teaches data collection of claim 5. Tsuyunashi further teaches, **wherein the authorization value corresponds to at least one of**:

- **a requesting entity for the vehicle policy data value; a vehicle type value; or a data type value of the collected vehicle data**; (FIG. 8 and FIG. 9, "The "request ID" is an identifier for identifying a collection request. The "collection conditions" are information indicating collection conditions for real data. The "collection conditions" include "object vehicle condition", "recording trigger", and "collection contents", for example. The "object vehicle condition" indicates a condition for vehicles serving as collection objects. The "recording trigger" indicates a trigger to start recording the real data in the on-vehicle device 50A. The "recording contents" are information indicating the real data recorded in the on-vehicle device 50A. In the example illustrated in FIG. 9, the object vehicles corresponding to the request ID "001" are vehicles "manufactured by XX

company", the recording trigger is acceleration (>XX G), and the collection contents are the positional information and the acceleration (in three seconds before and after the recording trigger)." [0129-0131]), where along with the user ID, information such as a vehicle type or a data type is included.

With respect to claim 7, Tsuyunashi discloses the limitations of claim 1. Tsuyunashi teaches data collection of claim 1. Tsuyunashi further teaches:

- **wherein the external event trigger condition comprises a location value, and wherein the policy execution circuit is further configured to collect vehicle data in response to the location value**; ("Alternatively, the present embodiment may employ a method of specifying a vehicle condition, that is, a method of collecting the real data R of the object vehicle by specifying a condition, such as vehicle model, traveling position (area), traveling time (time period), and vehicle with a specific event occurring, for example." [0084]), where the location can be used as an external event trigger.

With respect to claim 13, Tsuyunashi discloses the limitations of claim 1. Tsuyunashi teaches data collection of claim 1. Tsuyunashi further teaches:

- **wherein the external event trigger condition comprises a mission type value, and wherein the policy execution circuit is further configured to collect vehicle data in response to the mission type value**; ("The essential region R2 stores therein the collection conditions on which collection of data is essential. The essential region R2, for example, stores therein the collection conditions relating to human life. Specifically, the essential region R2 stores therein the collection conditions for emergency vehicles, such as fire engines and police vehicles." [0186], "If a fire or a traffic accident occurs near the vehicles, for example, the data collection apparatus 1A transmits

a transmission request for requesting camera video from the on-vehicle device 50A positioned near the scene. The data collection apparatus 1A provides the camera video to the emergency vehicles, thereby enabling the emergency vehicles to check the state of the scene before arriving at the scene." [0187]), where the mission type value includes first responder missions.

With respect to claim 14, Tsuyunashi teaches:

- **a request interface configured to interpret at least one vehicle data collection description and a target vehicle description**; ("The data user sets a data collection condition (user data collection request) using the user terminal 10 connected to the data collection apparatus 1. The data collection apparatus 1 creates tag data generation data for generating tag data having index-data properties. The tag data is associated with real data to be collected and is used to search for the data and grasp the outline of the data." [0046]), where a vehicle policy can be set, and therefore a data value such as the tag data including an external event trigger condition can be determined.
- **a policy creator circuit configured to determine at least one vehicle policy data value in response to the at least one vehicle data collection description and the target vehicle description**; ("The tag data generation data is generated based on an operation performed by the data user using a computer program and generation data stored in the user terminal 10 or the data collection apparatus 1. The data collection condition and the tag data generation data are stored in the data collection apparatus 1. The data collection condition and the tag data generation data are transmitted to data collection object vehicles (vehicle condition of which is specified by the data user) and also stored in the on-vehicle device 50." [0047]), where once a vehicle policy can be set, an external event trigger condition can be determined.

- **a policy implementation circuit configured to determine a group of vehicles in response to the target vehicle description, and to provide at least one of the at least one vehicle policy data values to the group of vehicles**; ("The "request ID" is an identifier for identifying a collection request. The "collection conditions" are information indicating collection conditions for real data. The "collection conditions" include "object vehicle condition", "recording trigger", and "collection contents", for example. The "object vehicle condition" indicates a condition for vehicles serving as collection objects. The "recording trigger" indicates a trigger to start recording the real data in the on-vehicle device 50A. The "recording contents" are information indicating the real data recorded in the on-vehicle device 50A. In the example illustrated in FIG. 9, the object vehicles corresponding to the request ID "001" are vehicles "manufactured by XX company", the recording trigger is acceleration (>XX G), and the collection contents are the positional information and the acceleration (in three seconds before and after the recording trigger)." [0129-0131]), which shows collecting information from a plurality of vehicles based on the specified information. FIG. 1C shows that the data is collected from a plurality of vehicles.

a cloud interface configured to receive at least one of:

- **at least a portion of identified vehicle data in response to the at least one vehicle policy data value**; ("FIGS. 1A to 1C are diagrams for explaining operations of the data collection system. As illustrated in FIG. 1A, the data collection system includes a user terminal 10, a data collection apparatus (server) 1, and on-vehicle device 50-1, 50-2, 50-3, . . . (when the on-vehicle device are indicated as a whole, they are referred to as on-vehicle device 50). The user terminal 10 is used by a data user, such as a developer of an autonomous car. The data collection apparatus 1 is formed by cloud, for example." [0045]), where the cloud interface receives vehicle information.
- **an alert response value in response to the at least one vehicle policy data value**; (If the data user uses and connects the user terminal 10 to the data collection apparatus 1 to check a data

collection state and collect the real data R, information (selection tag information) based on the tag data T collected from the data collection apparatus 1 is displayed on the user terminal 10. The user terminal 10 displays an operating screen for performing a data collection instruction operation based on the tag data T." [0050]), where an alert or display can be generated in response to the vehicle policy data.

With respect to claim 15, Tsuyunashi discloses the limitations of claim 14. Tsuyunashi teaches data collection of claim 14. Tsuyunashi further teaches:

- **wherein the policy implementation circuit is further configured to determine the group of vehicles in response to a shared vehicle aspect**; ("The "request ID" is an identifier for identifying a collection request. The "collection conditions" are information indicating collection conditions for real data. The "collection conditions" include "object vehicle condition", "recording trigger", and "collection contents", for example. The "object vehicle condition" indicates a condition for vehicles serving as collection objects. The "recording trigger" indicates a trigger to start recording the real data in the on-vehicle device 50A. The "recording contents" are information indicating the real data recorded in the on-vehicle device 50A. In the example illustrated in FIG. 9, the object vehicles corresponding to the request ID "001" are vehicles "manufactured by XX company", the recording trigger is acceleration (>XX G), and the collection contents are the positional information and the acceleration (in three seconds before and after the recording trigger). In this case, if the on-vehicle device 50A detects that the acceleration exceeds XX G, the on-vehicle device 50A records data on the acceleration in three seconds before and after the timing when the acceleration exceeds XX G together with the positional information." [0129-0132]), which shows collecting information from a plurality of vehicles based on the specified information. The example shows that data can be recorded from a

plurality of vehicles that have a shared vehicle aspect. FIG. 1C shows that the data is collected from a plurality of vehicles.

With respect to claim 16, Tsuyunashi discloses the limitations of claim 14. Tsuyunashi teaches data collection of claim 14. Tsuyunashi further teaches:

- **wherein the shared vehicle aspect comprises at least one of: a vehicle configuration value, a vehicle component value, or a vehicle feature value**; "The "vehicle model information" is information on the model of the vehicle and includes the model name and the model year. The "onboard equipment" is information on the equipment of the vehicle and includes information on whether a camera is present and the type of the camera, for example." [0126], which indicates the type of equipment, which is included as a vehicle component, on the vehicle.

With respect to claim 17, Tsuyunashi discloses the limitations of claim 14. Tsuyunashi teaches data collection of claim 14. Tsuyunashi further teaches:

- **wherein the shared vehicle aspect comprises at least one of: an owner value, a manufacturer value, an original equipment manufacturer value, a fleet value, a body builder value, or a dealer value**; ("The "on-vehicle device ID" is an identifier for identifying an on-vehicle device 50A. The "owner information" is information on the owner of the vehicle provided with the on-vehicle device 50A. While the example in FIG. 8 illustrates the name of the owner as the owner information, the owner information may include sex, age, address, and occupation of the owner, for example." [0125]. "The "object vehicle condition" indicates a condition for vehicles serving as collection objects. The "recording trigger" indicates a trigger to start recording the real data in the on-vehicle device 50A. The "recording contents" are information indicating the real data recorded in the on-vehicle device 50A." [0131]), which shows the owner information

and manufacturer information as vehicle identification information that can be shared vehicle information.

Claim Rejections - 35 USC § 103

7. In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

8. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

9. The factual inquiries for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.
2. Ascertaining the differences between the prior art and the claims at issue.
3. Resolving the level of ordinary skill in the pertinent art.
4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

10. This application currently names joint inventors. In considering patentability of the claims the examiner presumes that the subject matter of the various claims was commonly owned as of the effective filing date of the claimed invention(s) absent any evidence to the contrary. Applicant is advised of the obligation under 37 CFR 1.56 to point out the inventor and effective filing dates of each claim that

was not commonly owned as of the effective filing date of the later invention in order for the examiner to consider the applicability of 35 U.S.C. 102(b)(2)(C) for any potential 35 U.S.C. 102(a)(2) prior art against the later invention.

11. **Claim(s) 3, 8, 11-12, and 18-19** is/are rejected under 35 U.S.C. 103 as being unpatentable over Tsuyunashi et al. (US 20200077292 A1) in view of Adams et al. (US 20170021764 A1).

Regarding claims 3, 8, 11-12, and 18-19:

With respect to claim 3, Tsuyunashi discloses the limitations of claim 1. Tsuyunashi teaches data collection of claim 1. Tsuyunashi does not teach, but Adams teaches:

- **wherein the external event trigger condition comprises a weather condition value**; ("Further, the vehicle may provide autonomous features where the vehicle is connected to weather data and based on the weather data moves into the garage, for example. Alternatively, the vehicle may move to a safer location based on the weather data, for example. In either situation, the vehicle may monitor the weather information, and upon receipt of information that requires movement, may turn itself on and move as appropriate to aid in protecting the vehicle. Such a feature may reduce premiums on comprehensive by avoiding hail damage and other types of damage that occur as a result of weather accidents." [0101]), where weather data can be transmitted, which when combined with the trigger condition of Tsuyunashi, can be used to collect other vehicle data.

It would have been obvious to one of ordinary skill in the art before the effective filling date of the instant application to have combined Tsuyunashi's trigger condition with Adams' weather data so ("upon receipt of information that requires movement, may turn itself on and move as appropriate to aid in protecting the vehicle. Such a feature may reduce premiums on comprehensive by avoiding hail damage and other types of damage that occur as a result of weather accidents." See Adams [0101], and "Assessing the use of a vehicle using sensors on the vehicle relative to the expected use of the vehicle

and providing feedback to control the vehicle to adapt the use to the expected use is described." [0005]), where determining vehicle and environmental information can be used to provide more information for the vehicle.

With respect to claim 8, Tsuyunashi discloses the limitations of claim 7. Tsuyunashi teaches data collection of claim 7. Tsuyunashi does not teach, but Adams teaches:

- **wherein the location value comprises a jurisdiction description**; ("The DPU 170, using a multivariate analysis, generates the initial driver profile based on the provided biographic information verses the factors stored in the database 176. Where allowable by law, one factor that may be used in generating the initial risk assessment is based on the zip code of the insured's home/garaging address." [0110], "As one example, where allowable by law, the DLRI may be categorized by zip code. After receiving telematics data from the telematics device of vehicle 140, the DPU 170 may be configured to compare the driving location, with the DLRI to determine the relative risk of the locations." [0167], "The relativity factor may be relative to other drivers within the same demographic, driving on the same or similar roads under the same or similar conditions, or to the posted speed limit, or driving regulations." [0287]), where when the location of the vehicle is determined, jurisdiction information such as driving regulations can be determined.

It would have been obvious to one of ordinary skill in the art before the effective filling date of the instant application to have combined Tsuyunashi's trigger condition with Adams' location data because ("Assessing the use of a vehicle using sensors on the vehicle relative to the expected use of the vehicle and providing feedback to control the vehicle to adapt the use to the expected use is described." [0005]), where determining vehicle and environmental information can be used to provide more information for the vehicle.

With respect to claim 11, Tsuyunashi discloses the limitations of claim 7. Tsuyunashi teaches data collection of claim 7. Tsuyunashi does not teach, but Adams teaches:

- **wherein the location value comprises a season value**; ("In one scenario, the system 100 may receive telematics data for a fixed time period. In this scenario, the RPU 160 may be configured to compensate for the limited duration of the telematics data using a seasonality factor. For example, if the telematics data is received from September-December, and the biographical information indicates one of the insured drivers attends college away from home, RPU 160 may be configured to use the seasonality factor to adjust the pricing information to account for the lack of information transmitted regarding that driver. Conversely, under the same scenario, if the readings were taken during the summer, when the student was home, the telematics data may be skewed the other way. Accordingly, the RPU 160 may use the seasonality factor to account for those differences." [0189]), where location information including the seasonal information can be determined.

It would have been obvious to one of ordinary skill in the art before the effective filling date of the instant application to have combined Tsuyunashi's trigger condition with Adams' location data because ("Assessing the use of a vehicle using sensors on the vehicle relative to the expected use of the vehicle and providing feedback to control the vehicle to adapt the use to the expected use is described." [0005]), where determining vehicle and environmental information can be used to provide more information for the vehicle.

With respect to claim 12, Tsuyunashi discloses the limitations of claim 7. Tsuyunashi teaches data collection of claim 7. Tsuyunashi does not teach, but Adams teaches:

- **wherein the location value comprises a settlement description**; ("As an example above, the direct risk exposure may comprise information based on the location risk, which may be affected by vehicle density, lighting, outdoor/indoor parking, storing a vehicle in a neighborhood with a high number of break-ins or thefts, storing a vehicle in areas with high numbers of uninsured drivers. The DPU 170 may be configured to communicate with external servers 190 that may provide detailed crime information for predetermined areas (e.g. 1 meter). Additionally, the DPU 170 may communicate with external servers to determine weather information and real time traffic density and pedestrian density." [118]), where location information including the settlement information like vehicle desnity can be determined. Adams [0165] also describes categorizing roads into categories such as "urban" which is based off an environmental assessment.

It would have been obvious to one of ordinary skill in the art before the effective filling date of the instant application to have combined Tsuyunashi's trigger condition with Adams' location data because ("Assessing the use of a vehicle using sensors on the vehicle relative to the expected use of the vehicle and providing feedback to control the vehicle to adapt the use to the expected use is described." [0005]), where determining vehicle and environmental information can be used to provide more information for the vehicle.

With respect to claim 18, Tsuyunashi discloses the limitations of claim 14. Tsuyunashi teaches data collection of claim 14. Tsuyunashi does not teach, but Adams teaches:

- **wherein the shared vehicle aspect comprises at least one of: a maintenance value, a diagnostic value, a recall value, or a fault code value**; ("In an embodiment, the operational component data may include data representative of at least one category of fuel and/or air metering, emission control, ignition system control, vehicle idle speed control, transmission

control, hybrid propulsion or battery information. In another embodiment, the operational component data may include data based upon at least one of on-board diagnostic fault codes, trouble codes, manufacturer codes, generic codes or vehicle specific codes." [0010]), where vehicle information can include fault codes.

It would have been obvious to one of ordinary skill in the art before the effective filling date of the instant application to have combined Tsuyunashi's trigger condition with Adams' vehicle data because ("the present disclosure relates to monitoring and predicting component maintenance before an actual component failure to maximize maintainability and operational status of a fleet of vehicles thereby avoiding a vehicle breakdown." See Adams [0002]), to therefore benefit the monitoring of vehicle health.

With respect to claim 19, Tsuyunashi discloses the limitations of claim 14. Tsuyunashi teaches data collection of claim 14. Tsuyunashi does not teach, but Adams teaches:

- **wherein the shared vehicle aspect comprises at least one of an associated entity type value or a vehicle type value**; ("The question displayed on web page 802 solicits information where the user is requested to enter vehicle type, year, make, model, and other information. The user is also requested to enter information as to how the vehicle is paid for, how the vehicle is used, whether there is anti-theft equipment, and where the vehicle is stored." [0072]), where vehicle information can include the vehicle type.

It would have been obvious to one of ordinary skill in the art before the effective filling date of the instant application to have combined Tsuyunashi's trigger condition with Adams' vehicle data because ("the present disclosure relates to monitoring and predicting component maintenance before an actual component failure to maximize maintainability and operational status of a fleet of vehicles thereby

avoiding a vehicle breakdown." See Adams [0002]), to therefore benefit the monitoring of vehicle health.

12. **Claim(s) 9-10** is/are rejected under 35 U.S.C. 103 as being unpatentable over Tsuyunashi et al. (US 20200077292 A1) in view of Kyes et al. (US 20190244445 A1).

Regarding claims 9-10:

With respect to claim 9, Tsuyunashi discloses the limitations of claim 7. Tsuyunashi teaches data collection of claim 7. Tsuyunashi does not teach, but Kyes teaches:

- **wherein the location value comprises an altitude value**; ("The inventors have thus recognized and appreciated that, in some embodiments, monitoring operating conditions of an operational component may aid in generating a reliable prediction of whether and/or when an operational component will fail, or aid in increasing reliability of such a prediction. Such operating conditions may include environmental conditions, such as conditions in which a vehicle is being operated, including climate or weather conditions (temperature, humidity, altitude, etc.)" [0103]), where altitude data can be monitored, which when combined with the trigger condition of Tsuyunashi, can be used to collect other vehicle data in such conditions.

It would have been obvious to one of ordinary skill in the art before the effective filling date of the instant application to have combined Tsuyunashi's trigger condition with Kyes' location data because ("A new capability for monitoring and predicting vehicle component replacement before actual component failure thereby maximizing maintainability and operational status for each vehicle in a fleet of vehicles may also be provided." See Kyes [0006]), where determining vehicle and environmental information can be used to provide more information for the vehicle, such as monitoring vehicle health.

With respect to claim 10, Tsuyunashi discloses the limitations of claim 7. Tsuyunashi teaches data collection of claim 7. Tsuyunashi does not teach, but Kyes teaches:

- **wherein the location value comprises a climate value**; ("The inventors have thus recognized and appreciated that, in some embodiments, monitoring operating conditions of an operational component may aid in generating a reliable prediction of whether and/or when an operational component will fail, or aid in increasing reliability of such a prediction. Such operating conditions may include environmental conditions, such as conditions in which a vehicle is being operated, including climate or weather conditions (temperature, humidity, altitude, etc.)" [0103]), where climate data can be monitored, which when combined with the trigger condition of Tsuyunashi, can be used to collect other vehicle data in such conditions.

It would have been obvious to one of ordinary skill in the art before the effective filling date of the instant application to have combined Tsuyunashi's trigger condition with Kyes' location data because ("A new capability for monitoring and predicting vehicle component replacement before actual component failure thereby maximizing maintainability and operational status for each vehicle in a fleet of vehicles may also be provided." See Kyes [0006]), where determining vehicle and environmental information can be used to provide more information for the vehicle, such as monitoring vehicle health.

Conclusion

13. The prior art made of record and not relied upon is considered pertinent to applicant's disclosure.

- **Gao et al. (US 20180052025 A1)** is pertinent because "The merged collection of logged data allows selective mining and statistical treatment of: operating regimes across vehicles; near continuous operational data on specific vehicles or classes of vehicles; identification of vehicles or time periods where measured operation is outside of typical domains; selective filtering of

data based on operational data parameters (e.g., fuel rate and rpm where intake air temperature was >120° F.); and associating vehicle engines with keys linked to external oil analysis allowing environmental and operating conditions between oil sampling to be incorporated in the analysis of data using sampling time as the tie point to logged vehicle data." [0047] which pertains to collecting vehicle data.

14. Any inquiry concerning this communication or earlier communications from the examiner should be directed to Christine N Huynh whose telephone number is (571)272-9980. The examiner can normally be reached Monday - Friday 8 am - 4 pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Aniss Chad can be reached at (571)270-3832. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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